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Evaluating language models as language models

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How good is this model, for what we do?

A new base model ships most weeks. What arrives with it is a scorecard.

- A language model is trained to do one thing: **predict text**.
- So the direct question is: how well does it predict *our* text?
- Instead we read MMLU-Pro, HellaSwag, GSM8K.

Llama-3.2-1B vs Qwen2.5-1.5B

HellaSwag	2.1 points apart
Adapted BPB on news	3.4% apart
GSM8K	55.3 points apart

Two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. Which number describes the model?

Two reasons the scorecard misleads

1. Benchmarks are public

- Written before the model existed, for other purposes.
- Vendors optimise for the ones they report.
- Items and subject matter can reach pre-training data.
- Stronger model, or saw more of the exam?
No reader can tell.

2. They score a model before it is adapted

- Teams adapt a base model to their data first.
- What matters is how good it *becomes*.
- **Gemma-4-12B**, of fifteen models: 9th on GSM8K, 8th on MMLU-Pro, 13th on raw loss ...
- ... and **the best fine-tuned system of all**.

Why loss is the right quantity

There is no universal ground truth for “better” — so pick the principled yardstick

- Every base model was trained to minimise cross-entropy on text. Held-out loss is **its own objective**, measured out of sample.
- Cross-entropy is a proper scoring rule. In expectation over the domain's text p :

$$H(p, q) = \underbrace{H(p)}_{\text{same for every model}} + \text{KL}(p \parallel q)$$

so a difference in loss *is* a difference in distance from the text's distribution.

- The one obstacle: each model counts in its own tokens. Divide by **bytes**.

Bits per byte is not one metric among many. It is the training objective, put on a common scale.

Taking that seriously: three requirements

Bytes

every tokenizer
on one scale

Fresh, decontaminated text

no candidate
has trained on it

Equal adaptation

the same procedure
for every candidate

Decontamination, three layers:

- **Time** — text dated after every model shipped (Reddit and Hacker News, most of arXiv).
- **Hygiene** — exact dedup, quality filter, MinHash near-dup removal, 80/10/10 split with no overlap.
- **Membership screens** — Min-K++, context sensitivity, perturbation; the union of anything any model flags is quarantined for all (news snapshot).

Equal adaptation: the procedure, and what we ran

The method

-hparams sweep, default

- Per-model Optuna search: TPE sampler over LR, batch, warm-up, weight decay, LoRA rank, α , dropout.
- Successive halving prunes weak trials from 50 steps.
- Scored on the *validation* split; test never touched.
- Same number of trials for every model — no time cap.
- Winner trained until validation BPB stops improving.

What these results use

-hparams manual

- A full search of 17 models up to 35B was beyond our compute.
- One hand-picked recipe: LoRA $r=16$, $\alpha=32$, lr 10^{-4} , batch 32, fixed 250 steps.
- One A100: **3.7 min** at 0.5B, 11.6 at 7.7B, 24.7 at 12B, 62.7 at 31B.
- Where our earlier search settled: rank 16 won for 8 of 11 models.

A time-capped search is a trap: small models finish more trials and get better tuned. In our first run the 1B models completed 17–18 trials, the two largest only 4.

Checking it without ground truth

Two independent yardsticks — neither is a public benchmark

Fine-tune them and see

- Do what a team does: fine-tune 15 models on a real news task (write the lede).
- ROUGE-L on 500 articles no measurement touched.
- Expensive — run once.

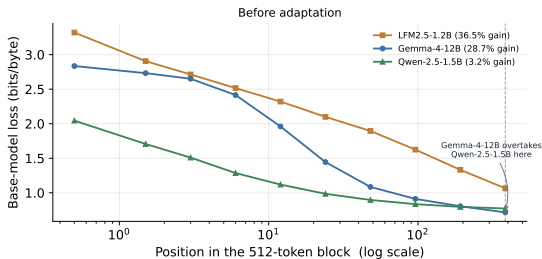
A quiz no model has seen

- Hide a name or number in a held-out article; the model writes it.
- Answered by the *released* weights, not the adapted ones.
- *Lenient*: answer starts in the first 24 characters (primary). *Strict*: exact start.

On news the two agree at Spearman **0.921** — so the cheap quiz stands in for the fine-tune on the other three corpora.

1. Adaptation changes the ordering

Eleven models, news quiz



- Adapted BPB: **7 of 11** in the exact quiz rank, mean error **0.36**.
- Unadapted: 2 of 11, mean error 2.00.
- **11 of 55** pairs flip wrong → right; none the other way.
- Why: unadapted loss is mostly *cold starts* — Gemma-4-12B opens a block at 2.83 BPB vs a 1.5B model's 2.04, and passes it by the end.

Zero-shot loss mixes *familiarity with this prose* into capability. A short equal adaptation removes the familiarity.

2. It recovers what benchmarks measure — and can be aimed

Rank correlation with each benchmark, eleven models

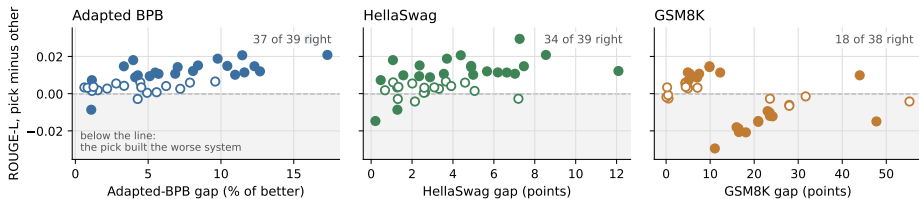
	Before adaptation	After adaptation
HellaSwag, on news / Reddit / Hacker News	0.63–0.72	0.96–0.98
GSM8K, adapted on arXiv mathematics	0.76	0.91
MMLU-Pro, adapted on arXiv mathematics	0.74	0.93

- HellaSwag is the benchmark closest to language modelling — and adapted BPB lands on it.
- Change the text to mathematics and the ranking swings to the reasoning benchmarks.

The choice of text points the ranking at the capability that matters.

3. It predicts what a model becomes

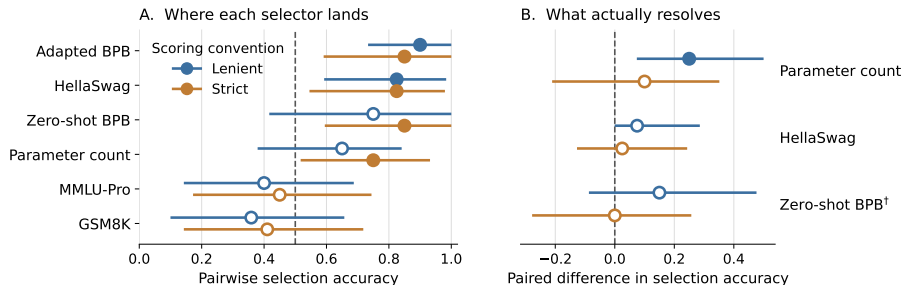
Fifteen models fine-tuned on news; 39 size-matched pairs



Adapted BPB, measured *before* fine-tuning: **0.949** of pairs right — HellaSwag 0.872, MMLU-Pro 0.513, GSM8K 0.474. **Every model within one rank** of where it finishes. Holds under a second, very different recipe (0.923).

4. It holds

Seventeen models, the 40 hardest (similar-size) pairs, and four corpora



Adapted BPB picks the better model **0.900** of the time (lenient), beats model size by $+0.250$ $[+0.074, +0.500]$, and on news, Reddit, Hacker News and arXiv math **never ranks below a public benchmark** under either scoring.

Using it

1. Candidates more than $\sim 2\times$ apart in size? The larger is usually better.
2. Otherwise: gather recent text from your domain, dated after every candidate shipped, and screen it.
3. Adapt every finalist by the same procedure — the per-model search where compute allows, the hand-picked recipe where it does not — and rank by adapted BPB.
4. Treat a gap under **2%** as a tie; then licence, latency or cost decide.
5. The zero-shot number is a free screen, not the selector.

No labels. No answer key. Text you own, that nobody could have trained on.

Evaluate language models as language models.

Questions?

Backup: per-model BPB, benchmark scores, contamination breakdown, longer-context transfer.

Backup: the adaptation results as a table

Lower BPB is better. Hand-picked recipe, 250 steps, news corpus.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.752	0.624	17.0%
Gemma-4-12B	Dense Transformer	12B	0.866	0.653	24.5%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.707	0.678	4.1%
Ministral-3-14B	Dense Transformer	14B	0.718	0.679	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.740	0.708	4.4%
Qwen-2.5-7B	Dense Transformer	7B	0.758	0.717	5.4%
Qwen-3.5-4B	Dense Transformer	4B	0.782	0.749	4.2%
Llama-3.2-1B	Dense Transformer	1.2B	0.811	0.771	4.9%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.833	0.797	4.3%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.340	0.852	36.4%
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.935	0.901	3.6%

Every value reproduces from the marker-masked fixed-protocol files in `results/domain_transfer/news_*.json`.

Backup: full benchmark scores

Eleven-model cohort, one host, one build

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624 (1)	57.1 (2)	87.9 (1)	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	61.4 (1)	85.6 (2)	86.6 (1)
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Percent, rank in parentheses; generated from the committed harness logs, the same table as the paper's. *lm-evaluation-harness* 0.4.12, bf16, greedy, native BOS; MMLU-Pro 5-shot CoT on a fixed 1,000-question subset, HellaSwag 10-shot length-normalised, GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

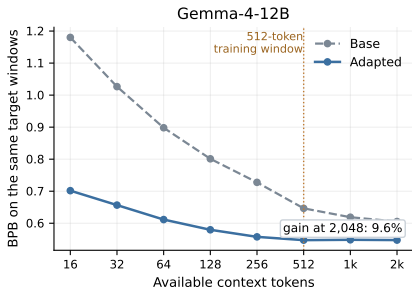
$N = 1000$ sampled sequences per model; 80 unique risk-flagged sequences removed

Model	Min-K++ flags	CoDeC flags	DCQ candidates	Preference	Risk flags
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Ministral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

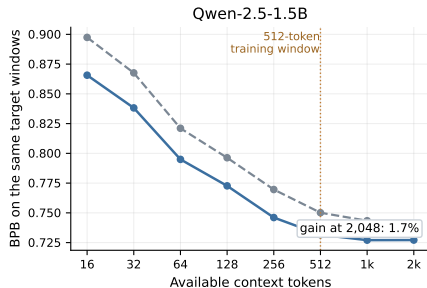
These are conservative quarantine signals on a preselected subset, not verified memorization rates. Exact overlap and model-likelihood signals must be interpreted separately.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes



Gemma base BPB keeps recovering after 512:
0.6465 → **0.6054** at 2,048 tokens.



Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.